LeakLens

Water Leak Localization System

Reading the Dashboard and How the Scenarios Work

Report 1 of 2 | B.Tech Project, IIIT Kottayam

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1 What this report is for

This report explains, in plain language, (i) every word and element you see on the LeakLens dashboard, (ii) what each kind of scenario really does to a water network, (iii) the formulas behind each decision, and (iv) exactly how the "Top-1" result is calculated and compared across scenarios. Every number is taken from the project's own result files and is measured on *held-out test scenarios* that the system never saw while learning (Section 6); all results are from simulation, not from real sensors.

2 The dashboard, piece by piece

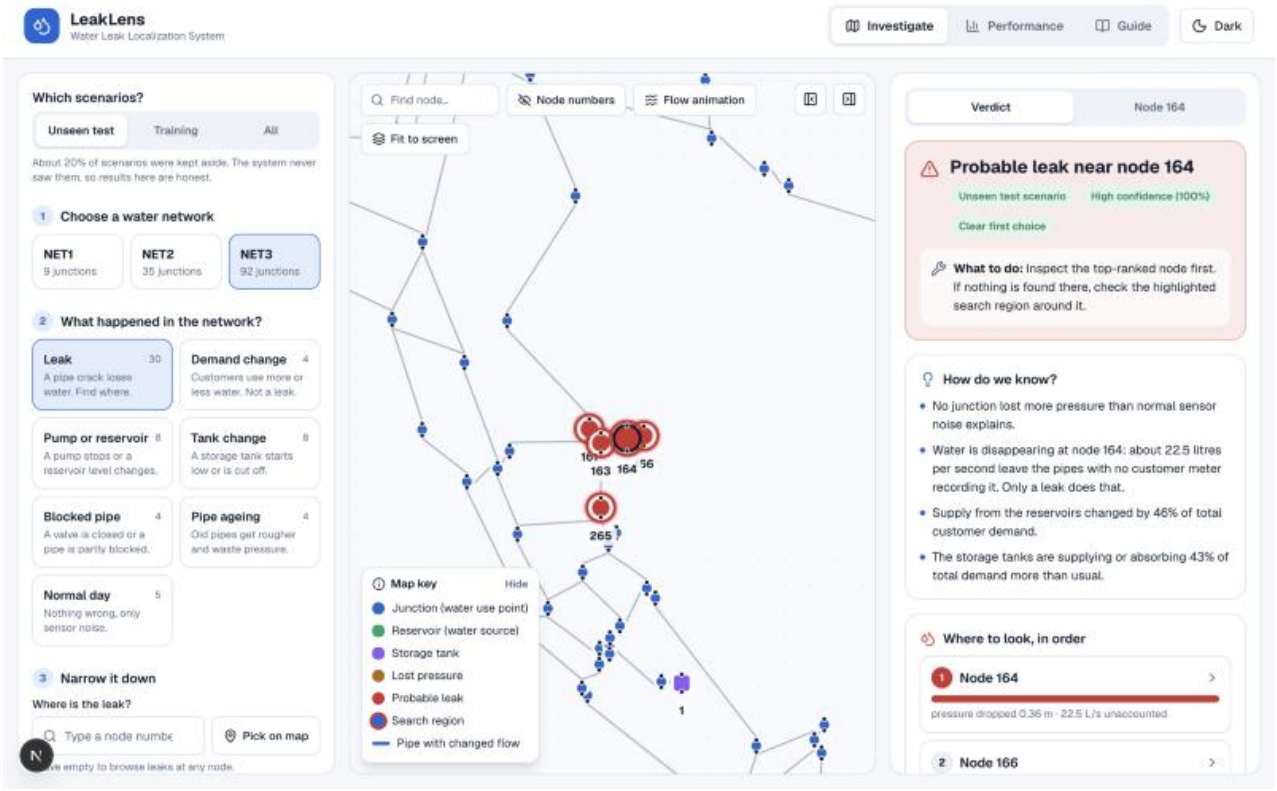


Figure 1: The Investigate screen after analysing a leak at node 115 (light theme). Left: choose a situation. Centre: network map. Right: verdict and reasons.

2.1 Header

- **LeakLens / Water Leak Localization System**: the project name and full title.
- **Investigate**: pick a scenario and look at its result on the map. **Performance**: how well the system works over all scenarios. **Guide**: a short help page with the glossary.

- **Light / Dark**: switches the colour theme. Your choice is remembered; the first time it follows your computer's setting.

2.2 Left panel: three steps

1. **Choose a water network.** NET1 (9 junctions, one pump, one tank), NET2 (35 junctions, one tank, no pumps), NET3 (92 junctions, 2 pumps, 3 tanks, 2 reservoirs). All three are standard EPANET benchmark networks.
2. **What happened in the network?** Seven kinds of situation: Leak, Demand change, Pump or reservoir, Tank change, Blocked pipe, Pipe ageing, Normal day. The small number on each card is how many scenarios of that kind exist for the chosen network.
3. **Narrow it down.** For leaks: *Where is the leak?* (type a node number or press *Pick on map*), *How big?* (Tiny, Small, Medium, Large, Burst) and *Conditions* (clean data, noisy sensors, hidden behind another event, starts at another time). The counter shows how many scenarios match; *Surprise me* runs a random one. Other kinds are searched by words such as "pump" or "30%".

Customers have smart demand meters tells the system whether the water used at each node is known. **Analyse this scenario** runs the analysis.

2.3 Map

Each circle is a **junction** (a point where customers draw water), squares are **reservoirs** (green, sources) and **tanks** (purple, storage). Lines are pipes. After an analysis: red circles are the predicted leak nodes (largest = first choice); orange circles lost pressure; red rings mark the **search region**; blue pipes carried a different flow than on a normal day. Click any node to see its pressure over the day.

2.4 Right panel: the verdict

Verdict box The cause found (for example "Probable leak near node 115") with a confidence label and a *What to do* sentence.

Confidence High if the model gives its answer at least 85% probability, Medium from 60%, otherwise Low. It is a model score, not a guarantee.

Clear first choice / Close call "Clear" when the second-ranked node scores less than half of the first; otherwise the first nodes are close and several should be checked.

How do we know? Plain-English reasons, built from the measurements: how many junctions lost pressure, whether water disappears without a meter noticing, whether a pump lost flow, and so on.

Where to look, in order The five highest-scoring nodes. The bar shows each node's score relative to the first.

Check against the answer key Because the data are simulated we know the true cause and the true leak node, so the dashboard shows whether the cause was right and whether Top-1 was hit (or how many pipes away it missed).

Technical details The probability of every cause, the 14 numbers the model looked at, and the score formula.

Node tab Pressure and unaccounted flow at the selected node, and a chart of pressure over 24 hours on the normal day versus this scenario.

2.5 Glossary of terms on screen

Term	Meaning
Pressure (m)	How hard water pushes in the pipe, written as metres of water column. Leaks and heavy use lower it.
Flow (L/s)	Litres of water moving through a pipe each second.
Baseline / normal day	The simulated day with nothing wrong. Every scenario is compared with it.
Sensor noise	Small random errors in real readings. The system learns their size so it does not raise alarms for them.
Unaccounted flow	Water leaving a node that no customer meter records. This is the leak flow.
Demand meters	Meters that report how much water customers draw at each node.
Leak score	Number used to rank nodes: unaccounted flow plus pressure drop, each scaled to 0–1.
Top-1 / Top-5	The real leak is the first node on the list / among the first five.
Pipe hops	Number of pipes between the predicted node and the real leak (0 = exact).
Search region	The predicted node and the nodes directly connected to it.
Confidence	How strongly the classifier prefers its answer over the other causes.

3 What a scenario is

A scenario is one 24-hour simulation of a network in which one thing (or a few things together) is changed from the normal day. The normal day is simulated first and called the *baseline*. Each scenario is stored with its true answer (which kind of event and, for leaks, which node), used only to score the system. The project contains 617 scenarios, split about 80% for learning and 20% for testing (Section 6).

Kind	Count	What is changed
Leak	299	A hole of area A opens at one junction at a given hour. Sizes: Tiny 0.0001, Small 0.0003, Medium 0.001, Large 0.003, Burst 0.01 m^2. Variants: every junction, five sizes, four start times (2, 8, 12, 20 h), sensor noise, and hidden behind another event.
Demand change	60	Water use at one node ×1.5 or ×0.5; all nodes ×1.3 or ×0.7; a hydrant opened for four hours.
Pump / reservoir	54	A pump switched off for 2, 4 or 6 hours while it normally runs; a reservoir head raised or lowered by 2, 5 or 10 m.
Tank change	60	A tank starts at 10–90% full, or is cut off from the network.
Blocked pipe	48	A busy pipe is closed at 10:00, or is partly blocked (large minor loss).
Pipe ageing	42	Every pipe becomes rougher (Hazen–Williams C multiplied by 0.6 to 0.95).
Normal day	54	Nothing changes; only measurement noise (small or large) or none at all.

Most non-leak scenarios also exist in a "noisy" copy, so the model sees realistic readings.

4 How each scenario changes the network

All scenarios are simulated by the EPANET engine through WNTR with pressure-dependent demand. Three tables are recorded for each hour: pressure at every node, flow in every pipe and the demand actually delivered at every node (what a customer meter would read).

4.1 Leak

Water escapes through an orifice. With discharge coefficient $C_d = 0.75$, leak area A, gravity g and gauge pressure head P at the node,

$$Q_\ell(t) = C_d\, A\, \sqrt{2\, g\, P(t)} \qquad (t \geq t_{\text{start}}). \tag{1}$$

The leak pulls extra water into that node, so flow rises in the pipes leading to it and pressure falls around it, most at the leak node. Because Q_ℓ grows with pressure, leaks at low-lying, high-pressure nodes lose more water. A leak is *local* (largest effect at one node) and *unmetered*.

4.2 Demand change

Customer use at node i is $D_i(t) = D_{\text{base},i}\, m(t)\, f$, where $m(t)$ is the daily pattern and f the scenario factor. More use pulls more flow through the pipes and lowers pressure nearby, like a leak, but the extra flow is *metered*: customers are drawing it.

4.3 Pump outage, reservoir head, tank level

These change the network's *boundary conditions*, the places where water and pressure enter. Switching a pump off removes the energy it adds, so pressure falls across everything it serves, tanks start to supply instead, and the pump's flow goes to zero. Raising or lowering a reservoir head by Δh shifts the pressure of the whole connected network by about Δh. A tank that starts low supplies less pressure and exchanges flow with the pipes. None of these lose water, and each affects *many* nodes at once.

4.4 Blocked pipe

Closing a pipe forces its flow elsewhere. Nodes downstream lose pressure, nodes upstream can gain it, and flows in neighbouring pipes change. No water is lost.

4.5 Pipe ageing

Older pipes are rougher. In the Hazen–Williams head-loss law

$$h_f = \frac{10.67\, L\, Q^{1.852}}{C^{1.852}\, d^{4.87}}, \tag{2}$$

a smaller roughness coefficient C means more pressure lost for the same flow. Pressure falls slowly and widely, and most where flow is highest. No water is lost.

4.6 What it looks like in the data

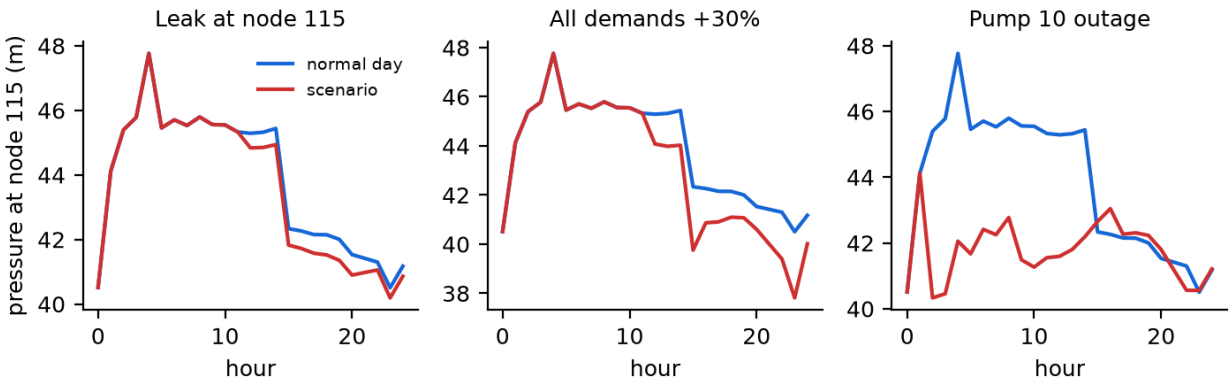


Figure 2: Pressure at NET3 node 115 on a normal day (blue) and in three scenarios (red). The leak causes a step down from hour 12; the demand surge lowers pressure all day; the pump outage drops it sharply while the pump is off.

Typical size of the effect, measured over all scenarios of each kind (median). *Drop* is the worst pressure drop at any junction; *Hit* is the share of junctions that lost more pressure than sensor noise explains; *Unacc.* is the largest unaccounted flow in multiples of its noise level.

Network	Kind	Scenarios	Drop (m)	Hit	Unacc.
NET1	Pipe ageing	14	9.97	100%	0.1
NET1	Demand change	20	1.00	72%	0.1
NET1	Leak	57	3.04	100%	22.4
NET1	Normal	18	0.09	0%	0.2
NET1	Pump / reservoir	18	0.28	0%	0.1
NET1	Tank change	12	9.91	100%	0.1
NET1	Blocked pipe	16	4.30	33%	0.1
NET2	Pipe ageing	14	0.29	0%	0.1
NET2	Demand change	20	0.11	0%	0.1
NET2	Leak	91	74.12	100%	14.6
NET2	Normal	18	0.10	0%	0.2
NET2	Tank change	12	0.57	50%	0.1
NET2	Blocked pipe	16	0.17	0%	0.1
NET3	Pipe ageing	14	5.84	98%	0.1
NET3	Demand change	20	0.14	0%	0.1
NET3	Leak	151	0.77	2%	7.9
NET3	Normal	18	0.13	0%	0.3
NET3	Pump / reservoir	36	1.92	1%	0.1
NET3	Tank change	36	2.49	32%	0.1
NET3	Blocked pipe	16	7.53	55%	0.1

Two things stand out: only leaks show a large unaccounted flow, and non-leak events either spread over most junctions (pumps, tanks, ageing, blockage) or are matched by customer meters (demand).

5 From measurements to a decision

5.1 Step 1: compare with the normal day

For node i and hour t the *pressure residual* is

$$r_i(t) = P_i^{\text{normal}}(t) - P_i^{\text{observed}}(t). \tag{3}$$

A reading only counts if the residual stays above the threshold for two consecutive hours (*persistence*), so one noisy reading cannot trigger anything. The peak persistent drop is $d_i = \max_t \min\{r_i(t), r_i(t{+}1)\}$.

5.2 Step 2: learn the noise

Each network has 18 normal runs containing only sensor noise; the learning (training) ones, about 14 per network, are used here. The pressure threshold is learned as

$$\tau_P = \max\{0.1,\ 4\,\sigma_P\}, \tag{4}$$

where σ_P is the standard deviation of those residuals ($\tau_P \approx 0.5$–0.6 m in all three networks). Flow and demand noise are learned in the same way. A fixed 0.5 m threshold, by contrast, raised an alarm on 41.7% of the normal test days.

5.3 Step 3: follow the water

Water cannot appear or vanish at a junction. For junction i, with pipe flows Q_p signed +1 if the pipe ends at i and −1 if it starts there,

$$\sum_p \text{sign}(p,i)\, Q_p(t) = D_i(t) + Q_{\ell,i}(t). \tag{5}$$

Subtracting the normal day and the change in metered demand leaves the leak:

$$u_i(t) = \Big[\sum_p \operatorname{sign}(p,i)\, \Delta Q_p(t) \Big] - \Delta D_i(t) \;=\; Q_{\ell,i}(t). \tag{6}$$

In noise-free data u_i is exactly the leak flow at node i and zero elsewhere. A demand change cancels in (6) because it is metered. Pressure and unaccounted flow are divided by their noise thresholds, so a value of 1 means "at the noise limit".

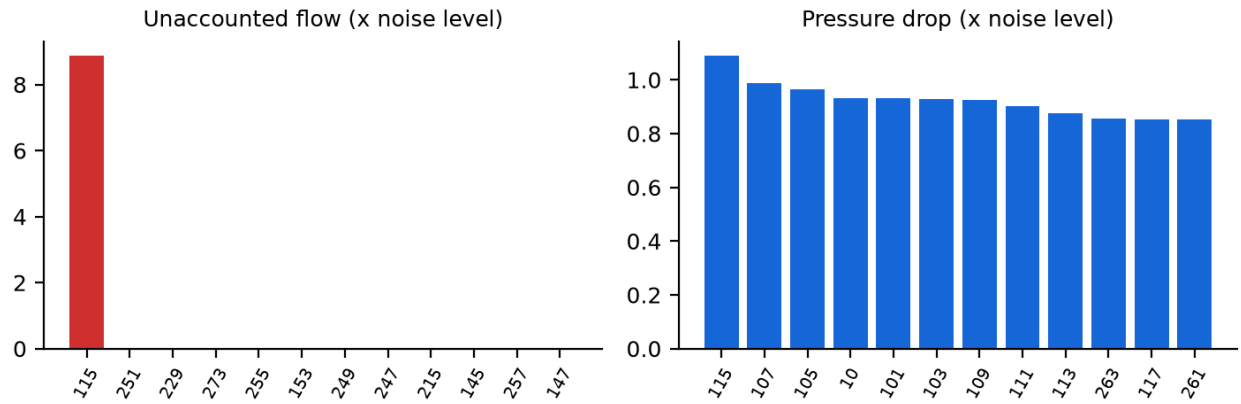


Figure 3: Leak at NET3 node 115. Left: unaccounted flow is large at node 115 and essentially zero elsewhere. Right: the pressure drop is high at 115 but also high at its neighbours, so pressure alone is a blurrier signal.

5.4 Step 4: fourteen summary numbers

Each scenario is summarised by 14 numbers, for example the share of junctions with a pressure drop, the largest unaccounted flow, the pump flow lost, and the change in reservoir supply and tank flow. Median values by kind of event:

Number	Normal	Leak	Demand	Pump/res.	Tank	Blocked	Ageing
Biggest pressure drop / noise threshold	0.20	4.25	1.11	2.61	4.29	6.82	4.28
Share of junctions with a pressure drop	0.00	0.56	0.03	0.01	0.82	0.44	0.96
Biggest unaccounted flow / noise	0.25	11.16	0.06	0.08	0.05	0.07	0.06
Share of nodes with changed metered demand	0.00	0.00	0.03	0.00	0.00	0.00	0.00
Pump flow lost (1 = off)	0.01	0.00	0.00	0.09	0.75	0.01	0.16
Reservoir supply change / total demand	0.01	0.01	0.01	0.37	0.46	0.01	0.34
Tank flow change / total demand	0.01	0.43	0.23	0.37	0.46	0.01	0.34
Biggest pressure rise / noise	0.20	0.54	0.44	2.62	3.20	3.29	6.28
Share of pipes with changed flow	0.00	0.60	0.35	0.47	0.60	0.65	0.81

5.5 Step 5: normal, or something else?

If every signal is within noise (all ratios below 1, almost no pipe flow change, no metered demand change) the answer is **normal**. This is the quiet gate; it is why normal days raise no false alarms, and also why very small leaks are reported as normal.

5.6 Step 6: which cause?

A random forest of 300 decision trees votes on the 14 numbers and outputs a probability for each of the six causes. The highest is the verdict; its probability gives the confidence label. Accuracy is measured on the held-out test scenarios only.

6 How the system is trained and tested

If a system is tested on the same scenarios it learned from, it will look perfect and we learn nothing. So all 617 scenarios were split once, with a fixed random seed, into about **80% for learning (478 scenarios)** and about **20% for testing (139 scenarios)**.

- **Learned from the training scenarios only:** the noise thresholds (from the normal training runs) and the cause classifier (the random forest).
- **Not learned at all:** the leak ranking of equation (7) is a fixed formula; its settings were chosen during development, before this split.
- **Measured on the test scenarios only:** every accuracy, false-alarm and Top-k figure in this report.
- **Near-copies stay together.** All leaks at the same junction (every size, start time, noise level and "hidden behind" variant) go to the same side, and a scenario and its sensor-noise twin go to the same side. Otherwise the test would be too easy.
- **Stratified.** Each network and each kind of event is split about 80/20; the test set has 12 scenarios of every non-leak kind and 67 leaks.
- **Dashboard tags.** Every scenario shows *Unseen test* or *Training*. Only the unseen ones give an honest check; the picker shows them by default.

Training versus test. Cause accuracy is 95.6% on the training scenarios and 82.7% on the unseen test scenarios, the gap you expect when a model has partly memorised; leak Top-1 is 97.8% on training leaks and 96.7% on test leaks, nearly the same, because the ranking is not trained.

7 How Top-1 is calculated

7.1 Scoring the nodes

Only if the verdict is "leak" are nodes ranked. For each junction i the two signals are scaled to 0–1 by dividing by their largest value in the network, $A_i = u_i^{\text{peak}} / \max_j u_j^{\text{peak}}$ and $D_i = d_i / \max_j d_j$, and combined:

$$S_i = A_i + 0.3\, D_i. \tag{7}$$

The unaccounted flow is the main signal; pressure drop is a backup that matters when flow data are noisy. If no node has an unaccounted flow above its noise limit, only the pressure term is used. Nodes are sorted by S_i from highest to lowest.

7.2 Worked example (NET3, leak at node 115)

The pressure threshold is 0.58 m. Node 115 has both the largest unaccounted flow and the largest pressure drop, so $A = 1.00$, $D = 1.00$ and $S = 1.00 + 0.3 \cdot 1.00 = 1.30$. Its neighbours have almost no unaccounted flow but a large pressure drop:

Rank	Node	A	D	Score S
1	115	1.00	1.00	1.30
2	107	0.00	0.91	0.27
3	105	0.00	0.89	0.27

The true node, 115, is ranked first, so this scenario counts as a Top-1 success: 0 pipe hops from the truth.

7.3 Rank, Top-k and error

For leak scenario s with true node n_s, let r_s be the position of n_s in the sorted list ($r_s = 1$ means first). Then

$$\text{Top-}k = \frac{1}{N}\sum_{s=1}^{N} \mathbf{1}[\, r_s \le k \,], \qquad \text{error}_s = \text{hops}\big(\hat{n}_s, n_s\big), \tag{8}$$

where $\hat{n}_s$ is the first-ranked node and hops is the number of pipes on the shortest path between the two nodes (the length in metres is also recorded). Over the 60 test leak scenarios scored, the true node was ranked first 58 times, so Top-1 = 58/60 = 96.7%. The first version of the system managed 35 of 60 (58.3%). With only 60 test leaks, one miss moves the percentage by about 1.7 points, so read small differences loosely.

7.4 Why Top-1 differs between scenarios

Method	Top-1	Top-3	Top-5	Mean hops
First version	58.3%	70.0%	71.7%	3.35
Final method	96.7%	98.3%	98.3%	0.13
Final + neighbour smoothing	96.7%	98.3%	98.3%	0.13
Final, no demand meters	93.3%	98.3%	98.3%	0.72
Pressure only	60.0%	70.0%	70.0%	3.05
Pressure only + smoothing	50.0%	63.3%	65.0%	3.18

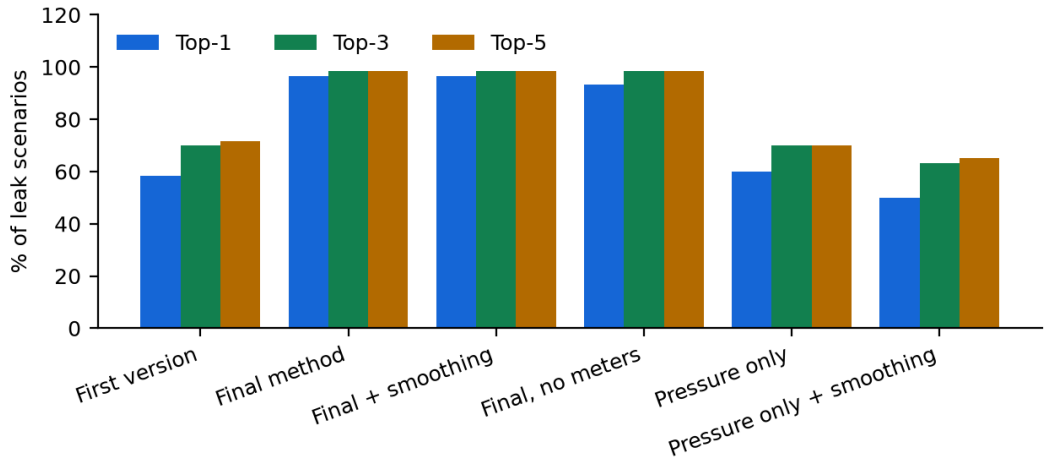


Figure 4: Localization accuracy of each method over all leak scenarios.

Reading the table: using unaccounted flow lifts Top-1 from 58.3% to 96.7%. Without demand meters Top-1 is 93.3%, because a demand change at another node can look like a small leak. Pressure alone gives only 60.0%, because the drop spreads to neighbours. Smoothing scores over neighbouring nodes gave 96.7%, no improvement, so the final method does not use it.

By scenario type, node type and network (final method, test leaks only; small groups):

Group	Leaks	Top-1	Top-5	Mean hops
Leak hidden behind another event	10	100.0%	100.0%	0.00
Leak with sensor noise	4	100.0%	100.0%	0.00
Leak size sweep	20	90.0%	95.0%	0.40
Leak start time	4	100.0%	100.0%	0.00
Leak at every junction	22	100.0%	100.0%	0.00
dead-end	17	100.0%	100.0%	0.00
hub	1	100.0%	100.0%	0.00
inline	42	95.2%	97.6%	0.19
net1	11	100.0%	100.0%	0.00
net2	26	92.3%	96.2%	0.31
net3	23	100.0%	100.0%	0.00

Why they differ:

- *Size.* Large leaks are easy to notice but the biggest (0.01 m^2) drops pressure across the whole network and can wipe out the unaccounted-flow signal (in the NET2 example the exact node was ranked 14th), so the exact node is harder to pick. Tiny leaks sit close to the sensor noise, so some are reported as normal and the dashboard then shows no ranking.
- *Hidden behind another event.* A hydrant or pump outage at the same time adds pressure changes elsewhere; the unaccounted-flow signal is unaffected when demands are metered, so these stay accurate.
- *Topology.* Hubs and dead ends behave differently; a hub's neighbours share its pressure drop, which is why pressure-only localization struggles there.
- *Network.* Net1 is tiny (9 junctions), so a "miss" is never far from the truth.

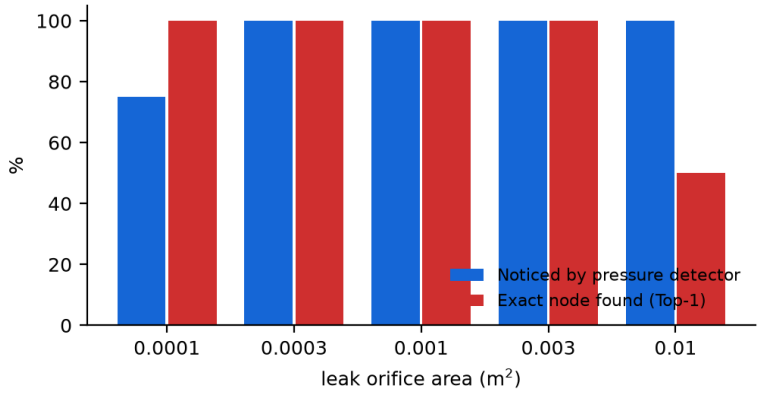


Figure 5: Effect of leak size on detection and on finding the exact node.

8 How the other verdicts are reached

- **Demand change**: pressure falls, but the unaccounted flow is near zero and the metered demand changed.
- **Pump or reservoir**: a pump's flow drops, or supply from the reservoirs changes, and most junctions lose pressure.
- **Tank change**: the tanks' net flow differs from normal.
- **Blocked pipe**: some junctions lose pressure while others gain it and many pipe flows change, with no unaccounted loss.
- **Pipe ageing**: a wide, smooth pressure loss with larger drops where flow is higher and no unaccounted loss.

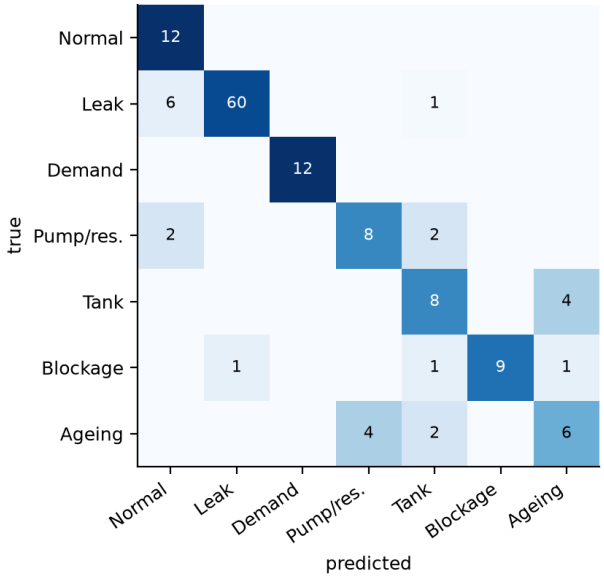


Figure 6: Which cause the system names for each true cause (counts, held-out test scenarios).

Leaks and demand changes are recognised very reliably. Pump, tank, blockage and ageing are sometimes confused with each other, because each makes the whole network lose pressure; and small events inside the noise are reported as normal.

9 What the Performance tab shows

It repeats these held-out test results in plain language, with the train-versus-test accuracy shown next to each other: leaks found at the exact node, leaks in the first five nodes, false alarms on normal days, right cause identified, the effect of leak size, the confusion between causes, results on a network the model has not seen (leave-one-network-out), and detailed per-group tables. The unseen-network results are weaker for fine-grained non-leak causes; this is the main work left for Semester 2.

10 Limits to keep in mind

- All data are simulated; real networks add noise and surprises.
- The test set is small (139 scenarios, 60 scored leaks), so every percentage has a wide margin of error.
- The leak-ranking settings and the 0.3 weight were chosen while looking at all scenarios, before the split was made; they were not re-tuned afterwards.
- Separating a leak from a demand change relies on smart demand meters at the nodes.
- Leaks that stay inside the sensor noise cannot be seen.
- Net3 nodes 61 and 601 and hour 21 are ignored because their pressure swings wildly for a tiny change (pump switching).